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GILGIT-BALTISTAN: Mountain Governance: Risks and Opportunities

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Understanding Pakistan in an Age of Geopolitical Transformation

SHAFQAT INSTITUTE FOR PAKISTAN AND EMERGING GEOPOLITICS(SIPEG)

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Key Takeaways

- Connectivity has transformed Gilgit-Baltistan from a remote frontier into a strategic corridor for trade, tourism, resource development and regional integration.
- The primary policy paradox is that infrastructure reduces the constraints of distance while increasing the economic consequences of climate- and hazard-related disruption.
- Mountain Governance must move beyond engineering and state penetration to integrate risk-informed planning, ecological stewardship, adaptive institutions and community participation.
- Tourism and mineral development can diversify livelihoods, but benefits need to be more equitably shared and supported by environmental safeguards, local ownership and value addition.
- Digital connectivity should be translated into digital livelihoods through access to education, telemedicine, e-commerce, remote work, public services and early-warning information.
- Territorial disputes and regional security competition make civilian infrastructure strategically sensitive; development, border management and conflict-risk reduction must be considered together.

Six Policy Priorities at a Glance

From connecting places to connecting people, knowledge and opportunities

01 MOUNTAIN RISK OBSERVATORY Integrate satellite, GIS, glacier, seismic and weather data with AI-assisted early warning.	**02 CLIMATE-RESILIENT KKH** Build hazard mapping, slope stabilization, monitoring, alternative routes and rapid response into corridor planning.
03 COMMUNITY-BASED TOURISM Prioritize local ownership, carrying capacity, conservation, waste and water management, and higher-value tourism.	**04 COMMUNITY-CENTERED MINERALS** Ensure transparent licensing, worker safety, environmental review, consultation, benefit-sharing and local processing.
05 DIGITAL LIVELIHOODS Convert fiber and mobile coverage into e-commerce, telemedicine, online education, remote work and disaster management.	**06 COMMON RESEARCH PLATFORM** Create a shared repository of publicly available research on Gilgit-Baltistan.

Glossary: Key Terms and Meanings

TERM	MEANING
AI-assisted early-warning systems	Tools that use artificial intelligence alongside monitoring data to identify emerging hazards and support timely alerts.
CPEC	China–Pakistan Economic Corridor: the connectivity and infrastructure framework discussed in the paper as a driver of regional integration.
Climate-resilient infrastructure	Infrastructure planned, designed, monitored and maintained to withstand or adapt to climate-related and geological hazards.
Community-based tourism	Tourism in which local communities retain meaningful ownership, participation and a share of the economic benefits, while protecting local environments and culture.
Digital livelihoods	Income-generating and service-access opportunities enabled by digital connectivity, including e-commerce, remote work, online education and digital services.
GIS	Geographical Information System: digital tools for collecting, mapping, analyzing and displaying geographically referenced information.
Glacial Lake Outburst Flood (GLOF)	A flood caused when water stored in a glacial lake is suddenly released, potentially affecting downstream settlements and infrastructure.
HKH	Hindu Kush Himalaya: the interconnected mountain region encompassing the Hindu Kush, Himalaya and Karakoram ranges in the paper's regional framing.
KKH	Karakoram Highway: the principal road corridor linking Gilgit-Baltistan with Pakistan's national road network and China.
Mountain Governance	The legal, institutional, technological, infrastructural and security processes through which states address the constraints of mountainous terrain and integrate remote territories into wider governance, markets and national strategy.

Remote sensing	The collection of information about the Earth's surface from a distance, commonly using satellites or aircraft.
Slope stabilization	Engineering and land-management measures intended to reduce the likelihood of slope failure, rockfall or landslides.
Value-added processing	Processing raw minerals or other resources locally so that more economic value, employment and related activity remain in the region.

Executive Summary:

Northern areas of Pakistan, at the confluence of the Hindu Kush Himalaya (HKH), has historically been defined by remoteness, rugged terrain and strategic location. The Karakoram Highway (KKH), expanding roads, tunnels, aviation, telecommunications and CPEC are gradually transforming this once-isolated frontier into a corridor of connectivity, infrastructure development, tourism, trade and geopolitical competition. Changing the dynamics of region's political economy. The emerging paradox is: the infrastructure that is "taming" the mountains is also intensifying exposure to climate change, glacial melt, landslides, flash floods and seismic hazards. The foremost policy challenge is therefore to move from connecting the HKH to governing HKH risks; balancing economic integration and technological innovation with ecological resilience, adaptive management, community participation, sustainable livelihoods, equitable development and prudent territorial disputes management.

Introduction:

Northern Areas of Pakistan, renamed in 2009 as Gilgit-Baltistan (GB), situated at the confluence of the Karakoram, Himalayas and Hindukush mountain ranges, has historically been defined by remoteness, difficult terrain and strategic location. The construction of the Karakoram Highway (KKH), followed by expanded road networks, tunnels, aviation, telecommunications and the China–Pakistan Economic Corridor (CPEC), has progressively transformed this relationship between *geography, state and society*. The region's political economy is undergoing transformation; once a remote frontier is increasingly becoming a corridor of connectivity, tourism, trade, resource extraction and geopolitical competition. This transformation can be understood as a process of **"**taming" the rugged mountain region, not through the conquest of nature, but through infrastructure, technology and institutions that weaken the constraints imposed by distance and topography. Concurrently, this process has generated an equally important new challenge: Climate change, glacier dynamics, landslides, flash floods and seismic risks increasingly threaten the infrastructure that has made mountain integration possible.

The policy challenge is; transiting from connecting the mountains to governing mountain risk. Tourism, mineral development and digital connectivity offer new opportunities for livelihood diversification, but they also raise questions of ecological sustainability, community ownership and fair distribution of benefits. The future governance strategy in GB, therefore, needs to combine connectivity with resilience, economic integration with environmental stewardship, adaptive management, and technological innovation with community participation. To understand dynamics of this socio-economic and geo-strategic change I will briefly analyze seven key factors

and suggest a few policy recommendations: (1) Mountain Governance: Taming the Karakoram, Himalayas and Hindukush (2) Infrastructural connectivity (3) Climate vulnerability & governance risks (4) Tourism and livelihoods (5) Mining and resource economy (6) Digital connectivity (7) Geopolitics of territorial disputes.

1. Mountain Governance: Taming the Karakoram, Himalayas and Hindukush

Gilgit-Baltistan was for much of its history geographically and politically peripheral to the territorial core of Pakistan. To borrow Tim Marshall's expression, it was 'prisoner of geography.'[1] Under British rule, the region functioned primarily as an imperial buffer between competing empires: Czarist Russia and the British and weak Qing dynasty China. After 1947, its difficult terrain, limited transport infrastructure and ambiguous constitutional status continued to constrain its integration into Pakistan.[2]

The construction of the Karakoram Highway between 1966 and 1978 marked a fundamental transformation (Had MDG's, SDG's Environmental laws, COP 31, existed in the late 1960's KKH, it would have been harder to build!!). The KKH physically connected GB with Pakistan's national road network and simultaneously created the foundation of Pakistan–China strategic cooperation. Roads, bridges, tunnels and later CPEC-related infrastructure progressively reduced the significance of physical distance. Today, GB has a population of 2 million in country of 245 million people and most towns of GB have road access throughout the year.[3]

This alteration represents more than infrastructure development. It constitutes what may be termed *Mountain Governance*: the legal, institutional, technological, infrastructural and security processes through which states curtail the political and economic constraints imposed by mountainous terrain and incorporate remote territories into wider systems of governance, markets, and national strategy.[4] The important analytical point is that geography has not disappeared as a constraint. Rather, *technology has modified the terms on which geography constrains society and*

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[1] Tim Marshall, *Prisoners of Geography: Ten Maps That Tell You Everything You Need to Know About Global Politics* (London: Elliott & Thompson, 2015), 20- 50.

[2] For an insightful historical overview of British rule, post-independence history, and the challenges of integration with the Pakistani state, see Yaqoob Bangash, "Three Forgotten Accessions: Gilgit, Hunza and Nagar," *The Journal of Imperial and Commonwealth History*, 38, no. 1 (2010), https://doi.org/10.1080/03086530903538269; For an extensively researched and wide-ranging study of GB, KHH, and connectivity in a globalized world, see Chad Haines, Nation, *Territory, and Globalization in Pakistan: Traversing the Margins* (London: Routledge, 2012). Chapters 1& 2.

[3] Pakistan-China Institute, "The Karakoram Highway: A Friendship Channel," Pakistan-China Institute, accessed September 8, 2026, www.pakistan-china.com. The highway was constructed through successive stages beginning in the 1960s and was completed in the late 1970s. For a wide-ranging historical, cultural, and security-related perspective, see Hasan H. Karrar, "The Geopolitics of Infrastructure and Securitization in a Postcolony Frontier Space," *Antipode* 54, no. 5 (2022): 1–21; Ayesha Siddique, *China-Pakistan Relations in the Twenty-First Century* (Lahore: Centre for Public Policy and Governance, 2022), 27–28.

[4] Mountain Research Initiative, "Mountain Governance," Active Working Groups, Community-Led Activities. https://mountainresearchinitiative.org/our-activities/community-led-activities/active-working-groups/mountain-governance

the state. The Karakoram frontier (i.e. GB) has consequently moved from inaccessibility towards integration.

2. Infrastructural Connectivity and the Limits of Engineering

The KKH remains the most visible symbol of this transformation. It is simultaneously an engineering marvel, a state-building instrument, a Pakistan–China strategic corridor and a livelihood artery for communities across northern Pakistan. Subsequent investment in the KKH, the Hazara–Thakot corridor, bridges, tunnels and associated CPEC infrastructure has deepened this connectivity.[5] The KKH links the unequal and remote parts of the country to its economic "cores", making them interdependent. That is what makes KKH a national identity marker and in this spirit the road is not simply a connectivity development project but also contributes towards solidifying a national identity. It memorializes and recognizes the sacrifices of Chinese and Pakistani engineers, workers and defense personnel who laid down their lives for building this highway—a monument of national importance.

Yet the history of the KKH also demonstrates the limits of engineering. Earthquakes, rockfalls, landslides, torrential rainfall, glacial activity and other natural hazards have repeatedly disrupted the road. In 1974, while the KKH was under construction an earthquake severely damaged the road and its re-construction was resumed after a month. Similarly, the 2005 earthquake caused extensive destruction, while the 2010 Attabad landslide submerged a major section of the KKH, blocked transport and created a large lake, requiring a substantial realignment of the road.[6] The resilience of the KKH is therefore as important as its engineering achievement. It has repeatedly been damaged, reconstructed and adapted, and continues to function as the principal lifeline connecting mountain communities with Pakistan's economic and administrative centers.[7] From my perspective, the KKH illustrates a broader transition: *the first generation of mountain governance sought to overcome the ruggedness of these mountain ranges; the next must learn to govern risk.*

3. Climate Vulnerability: From Taming HKH to Governing Risk

The urgency of this transition has been underscored by the catastrophic glacier-collapse disaster along the Nepal–China border in August 2026. A glacier collapse triggered an enormous flow of

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[5] Saba Shahid and Saeed Shafqat, *China Pakistan Economic Corridor: Demands, Dividends and Directions* (Lahore: Centre for Public Policy and Governance, 2018), 28–31. See also Haines, *Nation, Territory, and Globalization in Pakistan: Traversing the Margins*.

[6] Sher Sultan Baig, Garee Khan, and Mansoor Alam, "Geological Hazards Along the CPEC Route from Gilgit to Khunjerab," in *Silk Route Revisited: Essays and Perspectives on China Pakistan Economic Corridor and Beyond*, ed. Faqeer Muhammad, Saranjam Baig, et al. (Gilgit: China Study Center, Karakoram International University, 2023), 217–227. The 1974 earthquake was colossal; it affected almost 65 KM of the road under construction and caused heavy damage in Hunza, Hazara and Swat districts of Northern Pakistan, over 5000 died and several hundred were injured. It took several weeks to restore connection with the rest of the country.

[7] For an in-depth and wide-ranging analysis on politico- administrative challenges of governance and mining explorations, see Nadia Ali, "From *Khalisa Sarkar* Lands to Land Reforms: Legal Liminality, State Control and Resistance in Gilgit-Baltistan, Pakistan," *Third World Quarterly* 47, no. 3 (2026): 1–22, 10–11, https://doi.org/10.1080/01436597.2025.2573211. see also, Haines, *Nation, Territory, and Globalization in Pakistan: Traversing the Margins*. Chapters 3 and 4.

ice, rock and mud, destroying settlements and critical infrastructure around the Gyirong border region. By early September, more than 1,300 people had died and more than 5,000 remained missing. The disaster has deepened concerns across the HKH region about glacier instability, cascading hazards and the resilience of infrastructure built in high-risk mountain environments.[8] For Pakistan, the warning is particularly relevant. GB contains some of the world's most heavily glaciated mountain terrain outside the polar regions. Climate change is altering patterns of precipitation, glacier behavior, glacial lakes and slope stability. Pakistan's 7,000-plus glaciers, the largest collection outside the polar regions are melting rapidly, contributing to rising river levels, an increased risk of floods downstream (floods in Punjab & Sindh 2010, 2022 and heavy monsoons 2025), and more frequent glacial lake outbursts. The resulting hazards can rapidly affect roads, bridges, settlements, hydropower facilities and tourism infrastructure.

*The paradox is alarming: The more infrastructure connects the mountains, the greater the economic cost when climate-related disruption interrupts that connectivity***.** A landslide that once secluded a village can now disrupt a corridor connecting communities, markets, tourism destinations and cross-border trade.[9]

Mountain Governance must therefore incorporate climate and disaster risk into infrastructure planning from the outset. Satellite imagery, Geographical Information Systems (GIS), remote sensing, seismic monitoring, hydrological observation, glacier surveillance and AI-assisted early-warning systems should become integral to the management of the KKH and other critical infrastructure.[10] The policy objective should move from *infrastructure development to climate-resilient infrastructure development.*

4. Changing Dynamics of Political Economy: Integration, Tourism and Livelihoods Patterns

Infrastructural connectivity has opened GB to a rapidly expanding tourism economy. Hunza, Skardu, Khaplu, Naltar and other destinations have increasingly become part of Pakistan's domestic tourism circuit. Official GB data record almost *990,000 domestic tourist visits in 2024, with Skardu and Hunza among the principal destinations.* Tourism is significant not simply because it generates revenue, but because it is transforming the structure of local livelihoods. Traditional economies relied heavily on agriculture, livestock and remittances; now are increasingly complemented by hotels and guesthouses, restaurants, transport, guiding, handicrafts, adventure tourism, retail and digital services. Nadia Ali and Conrad Schetter have aptly described GB as the 'Tourism Frontier.'[11]

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[8] For a perceptive and informative analysis on the nature and scale of devastation, see Austin Lord, "A Cascading Disaster on the Nepal-China Border: What to Know About the August 2026 Rasuwa Flood, 72 Hours Later," *Stimson Center Commentary*, August 28, 2026, www.stimson.org/2026/a-cascading-disaster-on-the-china-nepal-border-what-to-know-about-the-august-2026-rasuwa-flood

[9] For an informative and insightful reporting and analysis, see Ali Tauqeer Sheikh, "Message from Nepal," *Dawn*, September 10, 2026, https://www.dawn.com/news/2028799.

[10] Saba Shahid and Saeed Shafqat, *China Pakistan Economic Corridor: Demands, Dividends and Directions* (Lahore: Centre for Public Policy and Governance, 2018)

[11] Nadia Ali and Conrad Schetter, "Tourism Frontier: A Space of Opportunity and Disenchantment in Hunza, Gilgit-Baltistan," *Political Geography* 127 (2026): 103514, https://doi.org/10.1016/j.polgeo.2026.103514. For an informative and entertaining read on tourism and its challenges, see Haines, *Nation, Territory, and Globalization in Pakistan: Traversing the Margins*. Chapter 4, "Emplacing the Karakoram Highway: from tourist spots to truck stops."

This shows a transition from relatively narrow subsistence economies towards diversified livelihood opportunities. Tourism can generate employment for youth, create opportunities for small enterprises and enable households to monetize cultural, environmental and recreational assets. It can also create new opportunities for women through hospitality, food processing, handicrafts and home-based enterprises.[12] But tourism also carries risks, road accidents, sudden landslides or stone falls cause many times loss of tourists and communities' lives. In addition, unregulated construction, waste generation, water supply pressure, land-price inflation and the commercialization of fragile landscapes can undermine precisely the environmental assets on which tourism depends. The challenge is not simply attracting more tourists, but to *develop community-based, high-value and environmentally sustainable tourism.* Tourism should become an instrument of local economic development rather than merely an external market imposed upon GB's communities.

5. Mining and the Emerging Resource Economy

Yet another significant transformation is taking place through mineral and gemstone development. GB possesses substantial deposits of precious and semi-precious gemstones, and artisanal and small-scale mining has long provided supplementary livelihoods in several communities. Research on Sumayar Valley demonstrates that gemstone mining has generated significant household income and enabled some families to invest in education, health, housing and information and communication technologies. At the same time, benefits are uneven, with higher-income households deriving greater livelihood gains, while informal and poorly regulated mining can impose environmental and occupational costs.[13] The crucial policy question is not how much mineral wealth GB possesses, *but how mineral wealth can be converted into broad-based local development*.

Like most of Pakistan, the region's mining economy remains characterized in many places by informal practices, limited safety standards, weak value chains and inadequate local processing. The absence of protective equipment and the occupational risks facing miners are not unusual. The future opportunity extends beyond gemstones. Growing international interest in critical minerals and rare-earth elements makes GB's geological resources strategically more meaningful. But resource development should not reproduce a simple extractive model in which minerals leave the region with little local value added. Equally important, local communities must participate in

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[12] For an in-depth and wide-ranging analysis on tourism, changing lifestyles, and mining explorations, see Nadia Ali, "From *Khalisa Sarkar* Lands to Land Reforms: Legal Liminality, State Control and Resistance in Gilgit-Baltistan, Pakistan," *Third World Quarterly* 47, no. 3 (2026): 1–22, 10–11, https://doi.org/10.1080/01436597.2025.2573211. For an equally exhilarating historical overview of Hunza in the 1930s and the author's lived experience in the early 1990s in Hunza, as well as the socio-economic transformation it has undergone, see Julie Flowerday, "Changes Over Time: Hunza in Treble Vision: 1930s and 1990s," in *Karakoram in Transition: Culture, Development, and Ecology in the Hunza Valley*, ed. Hermann Kreutzmann (Karachi: Oxford University Press, 2006), chap. 19, 273–284, 280–282.

[13] For informative, in-depth, and insightful analysis on this aspect, see Pascal Abb, "All Geopolitics Is Local: The China–Pakistan Economic Corridor Amidst Overlapping Centre–Periphery Relations," *Third World Quarterly* 44, no. 1 (2023): 76–95, https://doi.org/10.1080/01436597.2022.2128329. For further details, see Pascal Abb, Filippo Boni, and Hasan H. Karrar, *China, Pakistan and the Belt and Road Initiative: The Experience of an Early Adopter State* (New York: Routledge, 2024), 1–13. See also Hasan H. Karrar, footnote 3

decisions concerning access to resources, environmental protection and distribution of revenues. The existing customary community institutions do play an important role in governing access to mines. *Effective policy should therefore build upon, rather than simply displace, these local institutions.*

6. Digital Connectivity: From Connecting Places to Transforming Lives

The most significant transformation may ultimately come from digital rather than physical connectivity. Pakistan–China technological cooperation initially developed slowly after the 2001 institutional agreements. It gained momentum with CPEC after 2015 and increasingly shifted towards information and communication technologies, fiber infrastructure and institutional cooperation after 2020. The existing evidence of more than 1,000 kilometers of optical fiber deployment illustrates this transition.

For GB, digital connectivity has implications far beyond telecommunications. Fiber and mobile connectivity can allow mountain communities to access markets, education, telemedicine, digital banking, government services and information without overcoming the physical barriers of distance.[14] Today, a tourism operator in Hunza or Skardu can market accommodation internationally. A crafts producer can access customers beyond the region. Students can access online education. Health professionals can provide remote consultations. Farmers and traders can receive weather and market information. Most importantly, communities exposed to landslides, floods and glacier-related hazards can receive early warnings. How to make these equal opportunity accessible tools for the wider segment of people remains a challenge.

7. **Geopolitics of Territorial Disputes and Risk Management**

Geopolitically, Northern Areas (GB & Beyond) occupy one of the most sensitive mountain spaces in the world. The region combines the challenges of governing difficult terrain with the persistent uncertainties of unresolved territorial boundaries and competing security interests. Its strategic significance derives not only from its location at the intersection of the Karakoram, Himalayas and Hindukush, but also from its proximity to the contested frontiers of Pakistan, India and China, and to Afghanistan and the Central Asian states of Kyrgyzstan and Tajikistan. The history of conflict demonstrates how geopolitics becomes an overwhelming factor. The *Kargil conflict* of 1999 between India and Pakistan illustrated the military significance of high-altitude terrain, while the continuing *Siachen dispute* has produced one of the world's most stressful military deployments in a glaciated environment. To the east and northeast, the broader India–China boundary dispute has repeatedly transformed the Himalayan and Karakoram frontier into a zone of military confrontation, during the 1962 Sino-Indian War and again during the 2020 India–China military crisis. *These episodes demonstrate that mountains do not necessarily insulate states from conflict; rather, they can make territorial disputes more difficult to resolve while imposing extraordinary logistical and human costs on military operations.*

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[14] Saeed Shafqat and Ayesha Siddique, "China–Pakistan Technology and Artificial Intelligence (AI) Cooperation: Changing Dynamics of State & Private Sector Partnerships," *SIPEG Discussion Paper* (Lahore: Shafqat Institute for Pakistan and Emerging Geopolitics, January 2026), https://www.sipeg.org.

The topographic features of the region profoundly **shape the nature of warfare.** ***Roads, tunnels, bridges, airfields and telecommunications infrastructure therefore have a dual character: they enable civilian connectivity and economic development, but can also acquire strategic and military significance. The expanding infrastructure associated with the KKH and CPEC consequently has implications not only for trade and tourism but also for military mobility (troops deployment), surveillance, logistics (movement of heavy equipment) and border security. This creates a difficult governance dilemma: infrastructure designed to integrate mountain communities and promote development can simultaneously intensify the strategic competition surrounding the region. For example, another interesting development is that in early 2026, the Shaksgam Valley North of Siachen Glacier, a sparsely populated area, Chinese initiated infrastructure projects, roads, tunnels and bridges, that alarmed India, who recognize that China has superior capability in pursuing such projects. India considers it disputed territory, while China has actual control as a byproduct of China-Pakistan Boundary agreement 1963, and the territory is adjacent to Xinjiang and Tibet-Chinese sensitive security borders.***[15]

The wider geography further complicates this security environment. GB's connections toward Afghanistan, Kyrgyzstan and Tajikistan, place it within a larger Central Asian security and connectivity space, linking South Asia with China and the Eurasian interior. Border management therefore cannot be understood solely through the India–Pakistan or India–China lens. Transnational issues, including trade and transit, migration, terrorism, water and environmental security, disaster response and emerging digital connectivity add additional layers to the region's strategic significance.

Chad Haines has aptly characterized it as 'twin challenges of development and security' for GB. Consequently, the crucial challenge is: how *to govern a frontier where development, connectivity and security are intertwined*? The objective has to be: to ensure that infrastructure development, border management and civilian strength reinforce rather than undermine one another. For GB, sustainable peace and development will require greater capacity to manage disputed boundaries, difficult terrain and climate-induced hazards simultaneously. The region's future significance may depend on whether it can evolve from a historically contested frontier into **a *resilient zone of connectivity and cooperation, that curbs conflict escalation, without losing sight of the unresolved territorial and security realities that continue to shape it. Of course, it is easier said than done but must be brought to the attention of federal, provincial and local level decision makers.***

GB Governance: Six Policy Priorities

Le me restate, the first generation of infrastructure connected places. The emerging digital infrastructure can connect people, knowledge and opportunities. Based on the seven-factor analysis, I propose six policy priorities:

First, establish a Mountain Risk Observatory.
GB needs an integrated platform combining satellite imagery, Geographic Information System

[15] Daniel Markey, *The Shaksgam Valley in China's Borderlands: Mapping China's Strategic Space*, NBR Report (Seattle: The National Bureau of Asian Research, March 5, 2026).

(GIS), glacier monitoring, seismic information, weather data and AI-assisted early-warning systems. Expand on existing weather system collaborations with China on developing early warning information sharing and monitoring systems. Nepal disaster is a wake-up call that warns, Nepal, Bhutan, India, China, Bangladesh and Pakistan to either activate SARRC or focusing on climate matters seek regional cooperation on climate risks in HKH region. The six states may be encouraged to build consensus on, *one point Climate Risk Aversion Agenda* among the above six states; subsequently, Afghanistan, Kirghizstan and Tajikistan could join.

Second, make the KKH a climate-resilient corridor.

Future investment should incorporate slope stabilization, hazard mapping, alternative routes, emergency bridges, tunnels where appropriate, continuous monitoring and rapid-response mechanisms. To that effect develop integrated research on disaster risks and seek help from the University of Iceland, US Geological Survey, University of Michigan and may be a university from China, University of Iceland and the Karakoram University can take lead in such a venture.

Third, develop community-based tourism.
Tourism policy should prioritize local ownership, environmental carrying capacity, waste management, water conservation, cultural protection and higher-value rather than simply higher-volume tourism. Currently, more established communities like Hunza and Shigar derive more benefits from tourism, while other communities are trying to catch up. The sector requires additional research to understand the dynamics of local and international tourism in GB and how to harness the benefits more equitably.

Fourth, create a community-centered mineral development framework.
Mining regulation should incorporate transparent licensing, occupational safety, environmental assessment, community consultation, benefit-sharing, local employment and value-added processing within GB.

Fifth, move from digital connectivity to digital livelihoods.
Besides, fiber coverage, the federal government should help GB government to invest in digital entrepreneurship, e-commerce, telemedicine, online education, remote work and technology-enabled disaster management.

Sixth, establish a common research platform
The region attracts researchers and NGOs from all over the country and world. It would be in great interest, both academically and policymaking, to establish a common repository or database of all publicly available research work done in GB.

Conclusion:

Gilgit-Baltistan demonstrates that geography is not fate, but neither can geography be engineered away or be completely free from territorial disputes. Roads, tunnels, telecommunications, tourism and resource development have transformed the relationship between the Karakoram, Himalayas and Hindukush and the Pakistani state, its politics and economy. The region has changed from

isolation towards connectivity, and from *peripheral frontier to a strategic corridor.* On the other hand, this transformation has created a new policy challenge. Climate change and geological hazards are exposing the vulnerability of the very infrastructure that has made integration possible. The next generation of Mountain Governance must therefore move beyond the goal of simply ***taming geography***. Its objective should be to build a resilient relationship among ***mountains, technology, markets, state institutions, communities and neighboring states***. The, ultimate measure of success will not be how effectively the state penetrates the mountains, nor how many roads, mines or tourists the region can accommodate. It will be whether infrastructure and technology enable mountain communities to participate meaningfully in, and benefit equitably from, their integration into Pakistan and the wider regional economy, while preserving the ecological systems upon which their future depends.

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